Karnatak University's
Karnatak Arts/Science College, Dharwad

CLASS NOTES

B.Sc III Semester-Revised NEP
Algebra - II and Real Analysis - I
Unit - I: Groups

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Preface

These notes have been prepared for undergraduate B.Sc. students as an introduction to Group Theory. The aim is to explain the basic concepts in simple language, supported by examples and step-by-step proofs with reasons.

The notes cover groups, subgroups, permutations, cyclic groups, cosets, Lagranges theorem, normal subgroups, quotient groups, homomorphisms and isomorphisms. Students are encouraged to understand the definitions, work through the examples and attempt the proofs independently.

These notes are intended to support classroom learning and should be read alongside the recommended textbooks. Suggestions and corrections are welcome.

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1 Introduction

Preliminaries for Group Theory

Before studying group theory, we recall a few basic ideas about sets, functions and operations.

Sets and notation

A **set** is a well defined collection of distinct objects. The objects are called its **elements**.

We write $a \in A$ if a belongs to A, and $a \notin A$ if it does not.

Example. If $A = \{1, 2, 3\}$, then $2 \in A$ and $5 \notin A$. The order in which the elements are listed does not matter:

$$\{1, 2, 3\} = \{3, 1, 2\}.$$

Some frequently used sets are

$$\begin{aligned}
\mathbb{Z} &= \{\ldots, -2, -1, 0, 1, 2, \ldots\} = \text{the set of integers,} \\
\mathbb{Q} &= \left\{ \frac{p}{q} : p, q \in \mathbb{Z},\ q \neq 0 \right\} = \text{the set of rational numbers,} \\
\mathbb{R} &= \text{the set of real numbers,} \\
\mathbb{C} &= \text{the set of complex numbers.}
\end{aligned}$$

We also use

$$\mathbb{R}^+ = \{x \in \mathbb{R} : x > 0\}.$$

The empty set, denoted by $\varnothing$, has no elements.

Subsets and equality of sets

A set A is a **subset** of a set B, written $A \subseteq B$, if every element of A belongs to B.

Example.

$$\{1, 3\} \subseteq \{1, 2, 3\}.$$

Two sets are equal if they have exactly the same elements. To prove $A = B$, we often prove both

$$A \subseteq B \qquad \text{and} \qquad B \subseteq A.$$

Example.

$$\{2k : k \in \mathbb{Z}\} = \{\ldots, -4, -2, 0, 2, 4, \ldots\}.$$

Both descriptions represent the set of even integers.

Union, intersection and disjoint sets

For sets A and B,

$$\begin{aligned}
A \cup B &= \{x : x \in A \text{ or } x \in B\}, \\
A \cap B &= \{x : x \in A \text{ and } x \in B\}, \\
A \setminus B &= \{x : x \in A \text{ and } x \notin B\}.
\end{aligned}$$

Example. If $A = \{1, 2, 3\}$ and $B = \{3, 4\}$, then

$$A \cup B = \{1, 2, 3, 4\}, \qquad A \cap B = \{3\}, \qquad A \setminus B = \{1, 2\}.$$

The sets A and B are **disjoint** if $A \cap B = \varnothing$.

Example. The set of even integers and the set of odd integers are disjoint.

Ordered pairs and Cartesian products

An ordered pair (a, b) records two elements in a specified order. In general, $(a, b) \neq (b, a)$.

The **Cartesian product** of sets A and B is

$$A \times B = \{(a, b) : a \in A,\ b \in B\}.$$

Example. If $A = \{1, 2\}$ and $B = \{a, b\}$, then

$$A \times B = \{(1, a), (1, b), (2, a), (2, b)\}.$$

Functions, domain, codomain and image

A **function** $f : A \to B$ assigns to each element of A exactly one element of B.

Here A is the **domain**, B is the **codomain** and $f(A) = \{f(a) : a \in A\}$ is the **image** of f.

Example. For $f : \mathbb{Z} \to \mathbb{Z}$ defined by $f(n) = 2n$, the domain and codomain are both $\mathbb{Z}$, but the image is the set of even integers: $f(\mathbb{Z}) = 2\mathbb{Z}$.

One-to-one, onto and bijective functions

A function $f : A \to B$ is:

- **one-to-one** if $f(x) = f(y)$ implies $x = y$;
- **onto** if every element of B is the image of some element of A;
- **bijective** if it is both one-to-one and onto.

Example 1. The function $f : \mathbb{Z} \to \mathbb{Z}$ defined by $f(n) = 2n$ is one-to-one, since $2m = 2n$ implies $m = n$. It is not onto, since no integer maps to 1.

Example 2. The function $g : \mathbb{Z} \to \mathbb{Z}$ defined by $g(n) = n + 1$ is bijective. Indeed, $m + 1 = n + 1$ implies $m = n$, and every integer k is the image of $k - 1$.

Composition and inverse functions

If $f : A \to B$ and $g : B \to C$, their composition is $(g \circ f)(x) = g(f(x))$. Thus f is applied first and then g.

Example. Let $f(x) = x+1$ and $g(x) = 2x$ on $\mathbb{R}$. Then $(g \circ f)(x) = 2x+2$ and $(f \circ g)(x) = 2x + 1$. Thus the order of composition matters.

A bijective function $f : A \to B$ has an inverse function $f^{-1} : B \to A$ satisfying

$$f^{-1}(f(x)) = x \quad \text{and} \quad f(f^{-1}(y)) = y.$$

Example. If $f(x) = x + 1$, then $f^{-1}(x) = x - 1$. Here f^{-1} means the inverse function, not $1/f$.

Binary operations and closure

A **binary operation** on a nonempty set S is a function

$$* : S \times S \to S.$$

It assigns to each ordered pair (a, b) a unique element $a * b$ of the same set S.

The requirement that the result remains in S is called **closure**.

Example 1. Addition is a binary operation on $\mathbb{Z}$, since the sum of any two integers is an integer.

Example 2. Subtraction is not a binary operation on $\{0, 1, 2, \ldots\}$, since

$$2 - 5 = -3 \notin \{0, 1, 2, \ldots\}.$$

Associativity and commutativity

A binary operation $*$ on a set S is **associative** if

$$(a * b) * c = a * (b * c) \qquad \text{for all } a, b, c \in S.$$

It is **commutative** if

$$a * b = b * a \qquad \text{for all } a, b \in S.$$

Example 1. Integer addition is associative and commutative:

$$(2 + 3) + 4 = 2 + (3 + 4), \qquad 2 + 3 = 3 + 2.$$

Example 2. Integer subtraction is neither associative nor commutative:

$$(5 - 3) - 1 = 1, \qquad 5 - (3 - 1) = 3,$$

and

$$5 - 3 = 2, \qquad 3 - 5 = -2.$$

Remember. Associativity concerns the placement of brackets; commutativity concerns the order of the elements.

Identity elements and inverses

For a binary operation $*$ on S, an element $e \in S$ is an **identity element** if

$$e * a = a = a * e \qquad \text{for every } a \in S.$$

With respect to this identity, an element $b \in S$ is an **inverse** of a if

$$a * b = e = b * a.$$

Example 1: Addition. The identity for integer addition is 0, since

$$0 + a = a = a + 0.$$

The additive inverse of 5 is -5, because

$$5 + (-5) = 0 = (-5) + 5.$$

Example 2: Multiplication. The identity for real multiplication is 1. The multiplicative inverse of 2 is $\frac{1}{2}$, since

$$2 \cdot \frac{1}{2} = 1 = \frac{1}{2} \cdot 2.$$

However, $\frac{1}{2} \notin \mathbb{Z}$. Thus the existence of an inverse depends on the set as well as the operation.

Divisibility and the division algorithm

For integers d and n, with $d \neq 0$, we say d **divides** n, written $d \mid n$, if $n = dk$ for some integer k.

Example. $3 \mid 12$, since $12 = 3 \cdot 4$, but $3 \nmid 14$.

The **division algorithm** states that for every integer a and positive integer n, there exist unique integers q, r such that

$$a = qn + r, \qquad 0 \leq r < n.$$

Examples.

$$17 = 3 \cdot 5 + 2, \qquad -7 = (-2) \cdot 5 + 3.$$

Thus the remainders on division by 5 are 2 and 3, respectively. A remainder is always nonnegative.

The **greatest common divisor** of integers m, n, not both zero, is denoted by $\gcd(m, n)$. They are **relatively prime** if $\gcd(m, n) = 1$.

Examples.

$$\gcd(8, 12) = 4, \qquad \gcd(4, 9) = 1.$$

A brief recall of complex numbers

A complex number has the form $a + bi$, where $a, b \in \mathbb{R}$ and $i^2 = -1$. In particular,

$$i^0 = 1, \qquad i^1 = i, \qquad i^2 = -1, \qquad i^3 = -i, \qquad i^4 = 1.$$

Examples.

$$i(-i) = -i^2 = 1, \qquad (-1)(-i) = i.$$

These calculations will be useful when studying examples involving roots of unity.

Reading and writing mathematical statements

- **For every** means the statement must hold for all elements under consideration. To prove it, begin with an arbitrary element.
- **There exists** means at least one suitable element must be found.
- $P \Rightarrow Q$ means that P implies Q.
- $P \Longleftrightarrow Q$ means "P if and only if Q." Both directions must be proved.

Example of a direct proof. If m and n are even integers, write $m = 2r$ and $n = 2s$. Then

$$m + n = 2(r + s),$$

so $m + n$ is even.

Example of a counterexample. The statement "Every integer has a multiplicative inverse in $\mathbb{Z}$" is false: the inverse of 2 would be $\frac{1}{2}$, which is not an integer.

One counterexample is enough to disprove a statement that claims to hold for every element.

2 Unit-I: Groups

Definition 2.1 (Group)**.** A **group** is a nonempty set G together with a binary operation * such that, for all $a, b, c \in G$, the following axioms hold:

1. **Closure:** $a * b \in G$.
2. **Associativity:** $(a * b) * c = a * (b * c)$.
3. **Identity element:** There exists an element $e \in G$ such that $e * a = a * e = a$ for every $a \in G$.
4. **Inverse element:** For every $a \in G$, there exists an element $a^{-1} \in G$ such that $a * a^{-1} = a^{-1} * a = e$.

The group is denoted by $(G, *)$.
If, in addition,

5. $a * b = b * a$, for all $a, b \in G$,

then $(G, *)$ is called an **abelian group** or a **commutative group**.

(Question 1 & 2-QB)

Example 2.1.

1. The set $(\mathbb{Z}, +)$ of Integers is an abelian group under usual addition ' + '
2. The set $(\mathbb{Q}^\times = \mathbb{Q} \setminus \{0\}, \cdot)$ of non-zero Rational numbers is an abelian group under usual multiplication ' $\cdot$ '.
3. The set of Natural numbers $\mathbb{N} = \{1, 2, 3, \ldots\}$ under usual addition ' + 'is **not** a group.

(Question 1 & 2-QB)

Problem 2.1. Prove that the set $G = \{1, -1, i, -i\}$ of all fourth roots of unity forms an abelian group under multiplication.

(Question 25-QB)

Solution

Recall that $i^2 = -1$, $i^3 = -i$ and $i^4 = 1$. The Cayley table for multiplication is

$\cdot$	1	-1	i	$-i$
1	1	-1	i	$-i$
-1	-1	1	$-i$	i
i	i	$-i$	-1	1
$-i$	$-i$	i	1	-1

We verify the group axioms.

1. Closure: Every entry in the table belongs to G. Hence $ab \in G$ for all $a, b \in G$.
2. Associativity: Multiplication of complex numbers is associative. Since the elements of G are complex numbers, $(ab)c = a(bc)$ for all $a, b, c \in G$.
3. Identity: The element $1 \in G$ satisfies $1a = a = a1$ for every $a \in G$. Thus 1 is the identity element.

4. Inverse: We have $1 \cdot 1 = 1, \quad (-1)(-1) = 1, \quad i(-i) = 1 = (-i)i$.
   Therefore $1^{-1} = 1, \quad (-1)^{-1} = -1, \quad i^{-1} = -i, \quad (-i)^{-1} = i$. Thus every element of G has an inverse in G.
5. Commutativity: Multiplication of complex numbers is commutative. Hence

$$ab = ba \qquad \text{for all } a, b \in G.$$

   This is also evident from the symmetry of the Cayley table about its main diagonal.

Hence G is an abelian group under multiplication.

Problem 2.2. If $\mathbb{Q}_0$ is the set of all rational numbers other than zero and $*$ is a binary operation on $\mathbb{Q}_0$ defined by $a * b = \dfrac{ab}{5}$, $\forall a, b \in \mathbb{Q}_0$, then prove that $(\mathbb{Q}_0, *)$ forms an abelian group.

(Question 21-QB)

Solution

1. Closure:
   Let $a, b \in \mathbb{Q}_0$. Since a and b are nonzero rational numbers, $ab \neq 0$. Also, $\dfrac{ab}{5} \in \mathbb{Q}$ and $\dfrac{ab}{5} \neq 0$. Therefore, $a * b = \dfrac{ab}{5} \in \mathbb{Q}_0$.
   Hence, $\mathbb{Q}_0$ is closed under $*$.
2. Associativity:
   Let $a, b, c \in \mathbb{Q}_0$. Then $(a * b) * c = \left(\dfrac{ab}{5}\right) * c$.
   Using the definition of $*$, we have $(a * b) * c = \dfrac{\left(\frac{ab}{5}\right) c}{5} = \dfrac{abc}{25}$.
   Similarly, $a * (b * c) = a * \left(\dfrac{bc}{5}\right) = \dfrac{a\left(\frac{bc}{5}\right)}{5} = \dfrac{abc}{25}$.
   Thus, $(a * b) * c = a * (b * c)$.
   Therefore, $*$ is associative.
3. Identity:
   Let $e \in \mathbb{Q}_0$ be the identity element. Then $a * e = a$ for every $a \in \mathbb{Q}_0$.
   By the definition of $*$, we have $\dfrac{ae}{5} = a \Longrightarrow ae = 5a$.
   Since $a \neq 0$, $e = 5$.
   Also, $a * 5 = \dfrac{5a}{5} = a$ and $5 * a = \dfrac{5a}{5} = a$.
   Hence, the identity element is $\boxed{e = 5}$.
4. Inverse:
   Let $a \in \mathbb{Q}_0$. We seek $a^{-1} \in \mathbb{Q}_0$ such that
   $a * a^{-1} = 5$, where 5 is the identity element.
   Thus, $\dfrac{a\, a^{-1}}{5} = 5$.
   Therefore, $a\, a^{-1} = 25$, and hence $a^{-1} = \dfrac{25}{a}$.

Since $a \neq 0$, $\dfrac{25}{a} \in \mathbb{Q}_0$.

Moreover, $a * \dfrac{25}{a} = \dfrac{a\left(\frac{25}{a}\right)}{5} = \dfrac{25}{5} = 5$.

Similarly, $\dfrac{25}{a} * a = 5$.

Hence every element $a \in \mathbb{Q}_0$ has an inverse given by $\boxed{a^{-1} = \dfrac{25}{a}}$.

5. Commutativity:

For $a, b \in \mathbb{Q}_0$, it is given that $a * b = \dfrac{ab}{5}$.
Since ordinary multiplication of rational numbers is commutative, $ab = ba$.
Therefore, $a * b = \dfrac{ab}{5} = \dfrac{ba}{5} = b * a$.
Hence $*$ is commutative.
Therefore,

$$\boxed{(\mathbb{Q}_0, *) \text{ is an abelian group.}}$$

Problem 2.3. Show that the set $\mathbb{Q}_{-1}$ of all rational numbers except -1, with operation defined by $a * b = a + b + ab$, $\forall a, b \in \mathbb{Q}_{-1}$, is an abelian group.

(Question 22-QB)

Solution

1. Closure:
Let $a, b \in \mathbb{Q}_{-1}$. Since $a, b \in \mathbb{Q}$, $a + b + ab \in \mathbb{Q}$.
It remains to show that $a + b + ab \neq -1$.
Suppose, for contradiction, that $a + b + ab = -1$.
Then $ab + a + b + 1 = 0$.
Factoring, we get $(a + 1)(b + 1) = 0$.
Therefore, $a + 1 = 0$ or $b + 1 = 0$, which gives $a = -1$ or $b = -1$.
But $a, b \in \mathbb{Q}_{-1}$, so neither a nor b is equal to -1. This is a contradiction.
Hence, $a * b = a + b + ab \neq -1$.
Therefore, $a * b \in \mathbb{Q}_{-1}$.
Hence, $\mathbb{Q}_{-1}$ is closed under $*$.
2. Associativity:
Let $a, b, c \in \mathbb{Q}_{-1}$. Then $(a * b) * c = (a + b + ab) * c$.
Using the definition of $*$, we have $(a * b) * c = (a + b + ab) + c + (a + b + ab)c$.
Therefore, $(a * b) * c = a + b + c + ab + ac + bc + abc$.
Similarly, $a * (b * c) = a * (b + c + bc)$.
Using the definition of $*$, we have $a * (b * c) = a + (b + c + bc) + a(b + c + bc)$.
Therefore, $a*(b*c) = a+b+c+bc+ab+ac+abc = a+b+c+ab+ac+bc+abc$.
Thus, $(a * b) * c = a * (b * c)$.
Therefore, $*$ is associative.
3. Identity:

Let $e \in \mathbb{Q}_{-1}$ be the identity element. Then $a * e = a$ for every $a \in \mathbb{Q}_{-1}$.
By the definition of $*$, we have $a + e + ae = a$.
Therefore, $e + ae = 0$, or $e(1 + a) = 0$.
Since $a \neq -1$, $1 + a \neq 0$.
Hence, $e = 0$.
Also, $a * 0 = a + 0 + a(0) = a$ and $0 * a = 0 + a + 0(a) = a$.
Hence, the identity element is $\boxed{e = 0}$.

4. Inverse:
Let $a \in \mathbb{Q}_{-1}$. We seek $a^{-1} \in \mathbb{Q}_{-1}$ such that
$a * a^{-1} = 0$, where 0 is the identity element.
Thus, $a + a^{-1} + aa^{-1} = 0 \implies a^{-1}(1 + a) = -a$.
Since $a \neq -1$, $1 + a \neq 0$.
Hence, $a^{-1} = -\dfrac{a}{1+a}$.
Since $a \in \mathbb{Q}$ and $a \neq -1$, $-\dfrac{a}{1+a} \in \mathbb{Q}$.
We must also show that $-\dfrac{a}{1+a} \neq -1$.
Suppose $-\dfrac{a}{1+a} = -1$.
Then $-a = -(1 + a)$, which implies $-a = -1 - a$, and hence $0 = -1$, which is impossible.
Therefore, $-\dfrac{a}{1+a} \neq -1$.
Thus, $-\dfrac{a}{1+a} \in \mathbb{Q}_{-1}$.
Moreover, $a * \left(-\dfrac{a}{1+a}\right) = a - \dfrac{a}{1+a} - a\dfrac{a}{1+a}$.
Therefore, $a * \left(-\dfrac{a}{1+a}\right) = \dfrac{a(1+a) - a - a^2}{1+a} = 0$.
Similarly, $\left(-\dfrac{a}{1+a}\right) * a = 0$.
Hence every element $a \in \mathbb{Q}_{-1}$ has an inverse given by $\boxed{a^{-1} = -\dfrac{a}{1+a}}$.

5. Commutativity:
For $a, b \in \mathbb{Q}_{-1}$, it is given that $a * b = a + b + ab$.
Since ordinary addition and multiplication of rational numbers are commutative, we have $a + b = b + a$ and $ab = ba$.
Therefore, $a * b = a + b + ab = b + a + ba = b * a$.
Hence $*$ is commutative.
Therefore,

$$\boxed{(\mathbb{Q}_{-1}, *) \text{ is an abelian group.}}$$

General Properties of Groups

Note. For simplicity, throughout these notes we often write $\boxed{ab \text{ instead of } a * b}$. Here, ab denotes the group operation $*$ and does not necessarily represent ordinary multiplication.

Problem 2.4. Prove that identity element in a group is unique.
(Question 4-QB)

Solution

Suppose that e and e' are two identity elements of G. Then

$$\begin{aligned} e &= ee' && (\because\ e' \text{ is an identity element}),\\ ee' &= e' && (\because\ e \text{ is an identity element}). \end{aligned}$$

Hence $e = e'$. Therefore the identity element of a group is unique.

Problem 2.5. Prove that inverse of every element in a group is unique.
(Question 5-QB)

Solution

Let $a \in G$, and suppose that both b and c are inverses of a. Then by the inverse axiom $ab = ba = e$ and $ac = ca = e$. Now,

$$\begin{aligned} b &= be && (\because\ \text{identity axiom})\\ &= b(ac) && (\because\ ac = e, \text{ inverse axiom})\\ &= (ba)c && (\because\ \text{associativity})\\ &= ec && (\because\ ba = e, \text{ inverse axiom})\\ &= c && (\because\ \text{identity axiom}). \end{aligned}$$

Thus $b = c$. Hence the inverse of a is unique.

Problem 2.6. Prove that in a group $(G, *)$, $(a^{-1})^{-1} = a$, for every $a \in G$.
(Question 6-QB)

Solution

By the inverse axiom,

$$a^{-1}a = e.$$

Therefore,

$$\begin{aligned}
\left(a^{-1}\right)^{-1}\left(a^{-1}a\right) &= \left(a^{-1}\right)^{-1}e && (\because \text{ multiplying both sides on the left by } \left(a^{-1}\right)^{-1})\\
\left[\left(a^{-1}\right)^{-1}a^{-1}\right]a &= \left(a^{-1}\right)^{-1} && (\because \text{ associativity and identity axiom})\\
(e)a &= \left(a^{-1}\right)^{-1} && (\because \text{ since } \left(a^{-1}\right)^{-1} \text{ is the inverse of } a^{-1})\\
a &= \left(a^{-1}\right)^{-1} && (\because \text{ identity axiom}).
\end{aligned}$$

Hence,

$$\boxed{\left(a^{-1}\right)^{-1} = a}.$$

Problem 2.7. Let G be a group. Then for all $a, b \in G$, prove that $(ab)^{-1} = b^{-1}a^{-1}$. That is, prove that the inverse of a product is the product of the inverses taken in the reverse order.

(Question 28-QB)

Solution

Let $a, b \in G$. We show that $b^{-1}a^{-1}$ is both a right and a left inverse of ab.
First,

$$\begin{aligned}
(ab)(b^{-1}a^{-1}) &= a\left(bb^{-1}\right)a^{-1} && (\because \text{ associativity})\\
&= aea^{-1} && (\because\ bb^{-1} = e, \text{ inverse axiom})\\
&= aa^{-1} && (\because \text{ identity axiom})\\
&= e && (\because \text{ inverse axiom}).
\end{aligned}$$

Also,

$$\begin{aligned}
(b^{-1}a^{-1})(ab) &= b^{-1}\left(a^{-1}a\right)b && (\because \text{ associativity})\\
&= b^{-1}eb && (\because\ a^{-1}a = e, \text{ inverse axiom})\\
&= b^{-1}b && (\because \text{ identity axiom})\\
&= e && (\because \text{ inverse axiom}).
\end{aligned}$$

Hence $b^{-1}a^{-1}$ is the inverse of ab. By uniqueness of inverse,

$$\boxed{(ab)^{-1} = b^{-1}a^{-1}}.$$

Problem 2.8. Let G be a group and $a, b, c \in G$. Then prove that

$$ab = ac \implies b = c \quad \text{(left cancellation law)} \quad \text{and}$$
$$ba = ca \implies b = c \quad \text{(right cancellation law)}.$$

(Question 8-QB)

Solution

Suppose first that $ab = ac$. Multiplying both sides on the left by a^{-1}, we get

$$\begin{aligned} a^{-1}(ab) &= a^{-1}(ac) && (\because \text{ equality is preserved under left multiplication}) \\ (a^{-1}a)b &= (a^{-1}a)c && (\because \text{ associativity}) \\ eb &= ec && (\because\ a^{-1}a = e, \text{ inverse axiom}) \\ b &= c && (\because \text{ identity axiom}). \end{aligned}$$

Thus the left cancellation law holds.
Now suppose that $ba = ca$. Multiplying both sides on the right by a^{-1}, we obtain

$$\begin{aligned} (ba)a^{-1} &= (ca)a^{-1} && (\because \text{ equality is preserved under right multiplication}) \\ b(aa^{-1}) &= c(aa^{-1}) && (\because \text{ associativity}) \\ be &= ce && (\because\ aa^{-1} = e, \text{ inverse axiom}) \\ b &= c && (\because \text{ identity axiom}). \end{aligned}$$

Hence the right cancellation law also holds.

Problem 2.9. If $a, b \in G$, then prove that each of the equations $ax = b$ and $ya = b$ has a unique solution in G.

(Question 29-QB)

Solution

Consider first the equation $ax = b$. Since $a^{-1}, b \in G$, closure gives $a^{-1}b \in G$. Put $x = a^{-1}b$. Then

$$\begin{aligned} ax &= a(a^{-1}b) \\ &= (aa^{-1})b && (\because \text{ associativity}) \\ &= eb && (\because\ aa^{-1} = e, \text{ inverse axiom}) \\ &= b && (\because \text{ identity axiom}). \end{aligned}$$

Thus $x = a^{-1}b$ is a solution.
To prove uniqueness, suppose x_1 and x_2 are two solutions. Then $ax_1 = b = ax_2$, so $ax_1 = ax_2$. By the left cancellation law, $x_1 = x_2$.
Hence the solution is unique, and $\boxed{x = a^{-1}b}$.
Now consider $ya = b$. Since $b, a^{-1} \in G$, closure gives $ba^{-1} \in G$. Put $y = ba^{-1}$. Then

$$\begin{aligned} ya &= (ba^{-1})a \\ &= b(a^{-1}a) && (\because \text{ associativity}) \\ &= be && (\because\ a^{-1}a = e, \text{ inverse axiom}) \\ &= b && (\because \text{ identity axiom}). \end{aligned}$$

Thus $y = ba^{-1}$ is a solution.
If y_1 and y_2 are two solutions, then $y_1a = b = y_2a$, so $y_1a = y_2a$. By the right cancellation law, $y_1 = y_2$. Therefore, the solution is unique and $\boxed{y = ba^{-1}}$.

Other Problems

Problem 2.10. Prove that if $a^2 = e$ for every $a \in G$, then G is abelian. (**OR**)

Prove that in a group G, if every element has its own inverse, then G is abelian.(**OR**)

Prove that in a group G, if for every $a \in G$, $a^{-1} = a$, then G is abelian

(Question 10-QB)

Solution

For every $a \in G$, $a^2 = e \implies aa = e \implies a^{-1}(aa) = a^{-1}e \implies (a^{-1}a)a = a^{-1} \implies a = a^{-1}$. Hence a is its own inverse.
Similarly, $b^{-1} = b$ for every $b \in G$.
Now let $a, b \in G$. Since $ab \in G$ by closure, the hypothesis gives $(ab)^2 = e$.
Hence ab is its own inverse: $(ab)^{-1} = ab$. But,

$$\begin{aligned} (ab)^{-1} &= b^{-1}a^{-1} && (\because \text{ inverse of a product}) \\ &= ba && (\because\ a^{-1} = a \text{ and } b^{-1} = b). \end{aligned}$$

Therefore $ab = ba$ for all $a, b \in G$. Hence G is abelian.

Problem 2.11. Show that, for a group G, $(ab)^2 = a^2b^2$ for all $a, b \in G$ if and only if G is abelian.

(Question 7-QB)

Solution

Suppose first that G is abelian. Then $ba = ab$, and hence

$$\begin{aligned} (ab)^2 &= abab && (\because \text{ definition of square}) \\ &= aabb && (\because\ ba = ab, \text{ commutativity}) \\ &= a^2b^2 && (\because \text{ definition of powers}). \end{aligned}$$

Thus $(ab)^2 = a^2b^2$.
Conversely, suppose that $(ab)^2 = a^2b^2$ for all $a, b \in G$. Then

$$\begin{aligned} abab &= aabb && (\because \text{ expanding both sides}) \\ bab &= abb && (\because \text{ left cancellation by } a) \\ ba &= ab && (\because \text{ right cancellation by } b). \end{aligned}$$

Therefore $ab = ba$ for all $a, b \in G$, so G is abelian.

Problem 2.12. Prove that a group G is abelian if $b^{-1}a^{-1}ba = e$ for all $a, b \in G$.

Solution

Let $a, b \in G$. By hypothesis, $b^{-1}a^{-1}ba = e$. Multiplying both sides on the left by b, we get

$$\begin{aligned} b\left(b^{-1}a^{-1}ba\right) &= be \\ (bb^{-1})a^{-1}ba &= b && (\because \text{ associativity and identity axiom}) \\ a^{-1}ba &= b && (\because\ bb^{-1} = e, \text{ inverse axiom}). \end{aligned}$$

Now multiply both sides on the left by a:

$$\begin{aligned} a(a^{-1}ba) &= ab \\ (aa^{-1})ba &= ab && (\because \text{ associativity}) \\ ba &= ab && (\because\ aa^{-1} = e \text{ and identity axiom}). \end{aligned}$$

Thus $ab = ba$ for all $a, b \in G$. Hence G is abelian.

Problem 2.13. If G is a group of even order, prove that there exists an element $a \neq e$ such that $a^2 = e$.

Solution

Let $|G| = 2n$ for some positive integer n. We argue by contradiction. Assume that no element other than e satisfies $a = a^{-1}$.
The identity satisfies $e = e^{-1}$. For every non-identity element $a \in G$, our assumption gives $a \neq a^{-1}$.
Also, $a^{-1} \neq e$ whenever $a \neq e$. Therefore the $2n - 1$ non-identity elements can be grouped into disjoint pairs $\{a, a^{-1}\}$, each containing exactly two distinct elements.
Consequently, $2n - 1$ would have to be even. But $2n - 1$ is odd, a contradiction.
Hence there exists some $a \neq e$ such that $a = a^{-1}$.
Multiplying by a gives $a^2 = aa^{-1} = e$ by the inverse axiom. Thus there exists $a \neq e$ satisfying $a^2 = e$.

Subgroups

Definition 2.2 (Subgroup)**.** Let $(G, *)$ be a group. A non-empty subset H of G is called a **subgroup** of G if H itself forms a group under the same operation $*$ defined on G.

Example 2.2. Consider the group $(\mathbb{Z}, +)$ and let

$$H = 2\mathbb{Z} = \{\ldots, -6, -4, -2, 0, 2, 4, 6, \ldots\},$$

then show that H is a subgroup of $(\mathbb{Z}, +)$.

(Question 34-QB)

Solution

We show that H is a subgroup of $(\mathbb{Z}, +)$.

1. Closure:
   Let $a, b \in H$. Then $a = 2m, b = 2n$ for some $m, n \in \mathbb{Z}$. Hence $a + b = 2m + 2n = 2(m + n)$. Since $m + n \in \mathbb{Z}$, we have $a + b \in H$. Therefore, H is closed under addition.
2. Associativity:
   For $a, b, c \in H$, $(a + b) + c = a + (b + c)$, because addition in $\mathbb{Z}$ is associative.
3. Identity:
   Now, for every $a \in H$, $a + 0 = 0 + a = a$. Hence, 0 is the identity element in H.
4. Inverse:
   Let $a \in H$. Then $a = 2m$ for some $m \in \mathbb{Z}$. The additive inverse of a is $-a = -2m \in H$.

Hence $H = 2\mathbb{Z}$ is a subgroup of $(\mathbb{Z}, +)$.

Example 2.3. Consider the group $(\mathbb{R}\backslash\{0\}, \cdot)$ under multiplication. Let $H = \{1, -1\}$, then H is a subgroup of $(\mathbb{R} \setminus \{0\}, \cdot)$.

Example 2.4 (Non-example)**.** Consider the group $(\mathbb{Z}, +)$ and the subset

$$H = \{0, 1, 2, 3, \ldots\}.$$

Although H is closed under addition and contains the identity element 0, it does not contain the additive inverse of every element.
Hence H is **not** a subgroup of $(\mathbb{Z}, +)$.

Note. Trivial or Improper subgroups. Every group G, with identity element e, has the following subgroups:

1. **The subgroup $\{e\}$.** The set containing only the identity element forms a group under the operation of G, since:
   - It is closed: $e * e = e$.
   - Associativity is inherited from G.
   - e is its identity element.
   - The inverse of e is e itself.

   Hence, $\{e\} \subseteq G$ is a subgroup of G.
2. **The subgroup G.** Since every set is a subset of itself and G is already a group under its operation, it is a subgroup of itself.

A subgroup H is called a **non-trivial or proper subgroup** if $\{e\} \subsetneq H \subsetneq G$.

Example 2.5. In $(\mathbb{Z}, +)$, the identity element is 0. Thus, $\{0\}$ and $\mathbb{Z}$ are trivial or improper subgroups, while $2\mathbb{Z}$ is a non-trivial or proper subgroup because $\{0\} \subsetneq 2\mathbb{Z} \subsetneq \mathbb{Z}$.

Tests for a subgroup

The two-condition subgroup test

Theorem 2.1. *A nonempty subset H of a group G is a subgroup if and only if*
(i) $a, b \in H \implies ab \in H$ *and*
(ii) $a \in H \implies a^{-1} \in H$; *for all* $a, b \in G$.

(Question 30-QB)

Proof

Part-(i). Suppose that H is a subgroup of G. Closure axiom in H gives (i). Inverse axiom in H gives (ii).
Part-(ii). Suppose that (i) and (ii) hold. We verify the group axioms.
1. **Closure:** This is condition (i).
2. **Associativity:** For $a, b, c \in H \implies (ab)c, a(bc) \in H$ by (i) and $H \subseteq G \implies (ab)c = a(bc)$ because the operation is associative in G.
3. **Identity:** Let $h \in H$. By (ii), $h^{-1} \in H$, and by (i), $hh^{-1} \in H \implies hh^{-1} \in G$. Since $hh^{-1} = e$ in G, we obtain $e \in H$.
4. **Inverses:** This is condition (ii).

Thus H is a subgroup of G. □

Note (Two-condition subgroup test for additive group)**.** A nonempty subset H of an additive group $(G, +)$ is a subgroup of G if and only if:
(i) $a + b \in H$ for all $a, b \in H$;
(ii) $-a \in H$ for every $a \in H$.
Here $-a$ denotes the additive inverse of a in G.

The one-step subgroup test

Theorem 2.2. *A nonempty subset H of a group G is a subgroup if and only if*

$$a, b \in H \implies ab^{-1} \in H.$$

(KUD: Jan-2026)(Question 31-QB)

Proof

Part-(i). If H is a subgroup of G and $a, b \in H$, then $b^{-1} \in H$ by the inverse axiom in H, and $ab^{-1} \in H$ by closure axiom.
Part-(ii). Assume that $a, b \in H \implies ab^{-1} \in H$. We verify the group axioms.
1. **Identity:** For $h, h \in H$, applying the condition:
$$hh^{-1} = e \in H \quad (\because\ H \subseteq G \text{ and inverse axiom in } G\ ).$$
2. **Inverses:** For $a, e \in H$, applying the condition:
$$ea^{-1} \in H \implies a^{-1} \in H.$$

3. **Closure:** For $a, b^{-1} \in H$, applying the condition:

$$a(b^{-1})^{-1} \in H \implies ab \in H.$$

4. **Associativity:** For $a, b, c \in H \implies (ab)c, a(bc) \in H$ by closure axiom and $H \subseteq G \implies (ab)c = a(bc)$ because the operation is associative in G.

Thus H is a subgroup of G. □

Note (One-step subgroup test for additive group)**.** A nonempty subset H of an additive group $(G, +)$ is a subgroup of G if and only if $a - b \in H$ for all $a, b \in H$, where $a - b = a + (-b)$.

Example 2.6. Prove that the set of square roots of unity is a subgroup of the group of fourth roots of unity under multiplication.

(Question 33-QB)

Proof

Let $G = \{1, -1, i, -i\}$ be the group of fourth roots of unity under multiplication, and let $H = \{1, -1\}$ be the set of square roots of unity.
Clearly, $H \subseteq G$ and $H \neq \varnothing$.
Let $a, b \in H$. Since $1^{-1} = 1$ and $(-1)^{-1} = -1$, all possible values of ab^{-1} are:

a	b	ab^{-1}
1	1	1
1	−1	−1
−1	1	−1
−1	−1	1

Thus $ab^{-1} \in \{1, -1\} = H$ for every $a, b \in H$. Therefore, by the one-step subgroup test, H is a subgroup of G. □

Example 2.7 (**Example 2.2**)**.**

(Question 34-QB)

Solution

Clearly, $H = 2\mathbb{Z} \subseteq \mathbb{Z}$ and $H \neq \varnothing$.
Let $a, b \in H$. Then $a = 2m$ and $b = 2n$ for some $m, n \in \mathbb{Z}$.
Therefore, $a - b = 2m - 2n = 2(m - n) \in 2\mathbb{Z} = H$, since $m - n \in \mathbb{Z}$.
Thus $a - b \in H$ for every $a, b \in H$. Hence, by the one-step subgroup test for additive groups, H is a subgroup of $(\mathbb{Z}, +)$.

Intersection and union of subgroups

Intersection of two subgroups

Theorem 2.3. *If H_1 and H_2 are two subgroups of G, then $H_1 \cap H_2$ is also a subgroup G.*

(Question 32-QB)

Proof

Since both subgroups contain the identity of G, $e \in H_1 \cap H_2 \implies H_1 \cap H_2 \neq \emptyset$. If $a, b \in H_1 \cap H_2$, then $a, b \in H_1$ and $a, b \in H_2$. Because H_1 and H_2 are subgroups, their inverse and closure properties give $ab^{-1} \in H_1$ and $ab^{-1} \in H_2$. Hence $ab^{-1} \in H_1 \cap H_2 \implies H_1 \cap H_2$ is a subgroup of G. □

A union need not be a subgroup

Example 2.8. In $(\mathbb{Z}, +)$, consider

$$H_1 = 2\mathbb{Z} = \{\ldots, -6, -4, -2, 0, 2, 4, 6, \ldots\},$$
$$H_2 = 3\mathbb{Z} = \{\ldots, -12, -9, -6, -3, 0, 3, 6, 9, 12, \ldots\}.$$

For any positive integer d, the set $d\mathbb{Z}$ is a subgroup of $(\mathbb{Z}, +)$: it contains 0, and $dr - ds = d(r - s) \in d\mathbb{Z}$ for $r, s \in \mathbb{Z}$. This proves that H_1 and H_2 are subgroups using the one-step additive subgroup test.
However, $2, 3 \in H_1 \cup H_2$ while $2 + 3 = 5 \notin H_1 \cup H_2$, since 5 is divisible by neither 2 nor 3. Thus the union fails closure under addition and is not a subgroup.
But, if $H_3 = 4\mathbb{Z} = \{\ldots, -8, -4, 0, 4, 8, \ldots\}$, then $H_3 \subseteq H_1$, so $H_1 \cup H_3 = H_1$, which is a subgroup.

(Question 32-QB)

Exactly when is the union a subgroup?

Theorem 2.4. *If H_1 and H_2 are two subgroups of G, then $H_1 \cup H_2$ is a subgroup of G if and only if $H_1 \subseteq H_2$ or $H_2 \subseteq H_1$.*

Proof

Part-(i). If $H_1 \subseteq H_2$, then $H_1 \cup H_2 = H_2$, a subgroup. If $H_2 \subseteq H_1$, then $H_1 \cup H_2 = H_1$, also a subgroup.
Part-(ii). Suppose that $H_1 \cup H_2$ is a subgroup of G. Then we need to prove that $H_1 \subseteq H_2$ or $H_2 \subseteq H_1$.
Assume for a contradiction that $H_1 \not\subseteq H_2$ and $H_2 \subsetneq H_1$.
$H_1 \not\subseteq H_2 \implies a \in H_1$ and $a \notin H_2$. Similarly, $H_2 \not\subseteq H_1 \implies b \in H_2$ and $b \notin H_1$.
Since $a, b \in H_1 \cup H_2$, closure axiom gives $ab \in H_1 \cup H_2$.
There are two cases: $ab \in H_1$ or $ab \in H_2$.

Case 1: $ab \in H_1$**.** Since $a^{-1} \in H_1$, closure in H_1 gives $a^{-1}(ab) \in H_1$. But

$$\begin{aligned} a^{-1}(ab) &= (a^{-1}a)b && (\because \text{ associativity in the subgroup } H_1) \\ &= eb && (\because \text{ inverse axiom in the subgroup } H_1) \\ &= b && (\because \text{ identity axiom in the subgroup } H_1.) \end{aligned}$$

Thus $b \in H_1$, contradicting the choice of b.
Case 2: $ab \in H_2$**.** Since $b^{-1} \in H_2$, closure in H_2 gives $(ab)b^{-1} \in H_2$. But

$$\begin{aligned} (ab)b^{-1} &= a(bb^{-1}) && (\because \text{ associativity in the subgroup } H_2) \\ &= ae && (\because \text{ inverse axiom in the subgroup } H_2) \\ &= a && (\because \text{ identity axiom in the subgroup } H_2.) \end{aligned}$$

Thus $a \in H_2$, contradicting the choice of a. Both cases are impossible. Hence $H_1 \subseteq H_2$ or $H_2 \subseteq H_1$. □

Addition and Multiplication Modulo an Integer

Definition 2.3. Let n be a positive integer. For any integers a and b, the *addition of a and b modulo n*, denoted by $a +_n b$, is the least nonnegative remainder obtained when $a + b$ is divided by n.
☞ Thus, $a +_n b = r$, where $a + b = qn + r$ with $0 \leq r < n$, for some integer q.

Example 2.9.
1. $4 +_5 3 = 2$, since $4 + 3 = 7 = 1 \cdot 5 + 2$.
2. $5 +_6 1 = 0$, since $5 + 1 = 6 = 1 \cdot 6 + 0$.
3. $(-4) +_5 1 = 2$, since $-4 + 1 = -3 = (-1) \cdot 5 + 2$.

Theorem 2.5. *For every positive integer m, $(\mathbb{Z}_m = \{0, 1, \ldots, m-1\}, +_m)$ is a finite abelian group of order m.*

Problem 2.14. Show that $G = \{0, 1, 2, 3, 4\}$ is a finite abelian group under addition modulo 5.

(Question 26-QB)

Solution

Let $G = \{0, 1, 2, 3, 4\}$. The Cayley table (composition table) for addition modulo 5

is

$+_5$	0	1	2	3	4
0	0	1	2	3	4
1	1	2	3	4	0
2	2	3	4	0	1
3	3	4	0	1	2
4	4	0	1	2	3

1. **Closure.** Every entry in the table belongs to G. Hence $a +_5 b \in G$ for all $a, b \in G$.
2. **Associativity.** Let $a, b, c \in G$. Then by the division algorithm, $(a +_5 b) +_5 c$ is the remainder when $(a+b)+c$ is divided by 5 and $a +_5 (b +_5 c)$ is the remainder when $a + (b + c)$ is divided by 5.
   Hence $(a +_5 b) +_5 c = a +_5 (b +_5 c)$.
3. **Identity.** For every $a \in G$, $0 +_5 a = a = a +_5 0$. Thus 0 is the identity element.
4. **Inverses.** From the table, $0{+_5}0 = 0, \quad 1{+_5}4 = 0 = 4{+_5}1, \quad 2{+_5}3 = 0 = 3{+_5}2$. Hence the additive inverses of $0, 1, 2, 3, 4$ are $0, 4, 3, 2, 1$, respectively.
5. **Commutativity.** The Cayley table is symmetric about its main diagonal. Therefore, $a +_5 b = b +_5 a$ for all $a, b \in G$.

Hence $(G, +_5)$ is a finite abelian group of order 5.

Definition 2.4. Let n be a positive integer. For any integers a and b, the *multiplication of a and b modulo n*, denoted by $a \times_n b$, is the least nonnegative remainder obtained when ab is divided by n.

☞ Thus, $a \times_n b = r$, where $ab = qn + r$ with $0 \le r < n$, for some integer q.

Example 2.10.

1. $3 \times_5 4 = 2$, since $3 \cdot 4 = 12 = 2 \cdot 5 + 2$.
2. $2 \times_6 3 = 0$, since $2 \cdot 3 = 6 = 1 \cdot 6 + 0$.
3. $(-2) \times_5 3 = 4$, since $(-2) \cdot 3 = -6 = (-2) \cdot 5 + 4$.

Theorem 2.6. *Let p be prime and $G = \{1, 2, \ldots, p-1\}$. Then $(G, \times_p)$ is a finite abelian group of order $p - 1$.*

Problem 2.15. Show that $G = \{2, 4, 6, 8\}$ is a finite abelian group under multiplication modulo 10.

(Question 27-QB)

Solution

Let $G = \{2, 4, 6, 8\}$. The Cayley table (composition table) for multiplication modulo

10 is

$\times_{10}$	2	4	6	8
2	4	8	2	6
4	8	6	4	2
6	2	4	6	8
8	6	2	8	4

1. **Closure.** Every entry in the table belongs to G. Hence $a \times_{10} b \in G$ for all $a, b \in G$.
2. **Associativity.** Let $a, b, c \in G$. Then by the division algorithm, $(a \times_{10} b) \times_{10} c$ is the remainder when $(ab)c$ is divided by 10 and $a \times_{10} (b \times_{10} c)$ is the remainder when $a(bc)$ is divided by 10.
   Hence $(a \times_{10} b) \times_{10} c = a \times_{10} (b \times_{10} c)$.
3. **Identity.** From the row and column labelled 6,
   we have $6 \times_{10} a = a = a \times_{10} 6$ for every $a \in G$. Thus 6 is the identity element.
4. **Inverses.** From the table, $2 \times_{10} 8 = 6 = 8 \times_{10} 2, \quad 4 \times_{10} 4 = 6, \quad 6 \times_{10} 6 = 6$.
   Since the identity is 6, the inverses of $2, 4, 6, 8$ in G are $8, 4, 6, 2$, respectively.
5. **Commutativity.** The Cayley table is symmetric about its main diagonal. Therefore, $a \times_{10} b = b \times_{10} a$ for all $a, b \in G$.

Hence $(G, \times_{10})$ is a finite abelian group of order 4.

Cyclic Groups and Generators

Example 2.11 (Order of an Element). Let G be a group and $a \in G$. The **order of the element** a is the *smallest positive integer* n such that $a^n = e$, where e is the identity element of G.
It is denoted by $\boxed{o(a) = n}$.

- If no positive integer n satisfies $a^n = e$, then a is said to have **infinite order**.
- In additive notation, the order of an element $a \in G$ is the smallest positive integer n such that $na = \underbrace{a + a + \cdots + a}_{n \text{ times}} = 0$.

Example 2.12. In the multiplicative group $\{1, -1\}$ For $a = -1$, $(-1)^1 = -1 \neq 1$ and $(-1)^2 = 1$. Therefore, $\boxed{o(-1) = 2}$.

Example 2.13. In $(\mathbb{Z}_6, +)$. For $a = 2$, $2(2) = 2 +_6 2 = 4$ and $3(2) = 2 +_6 2 +_6 2 = 0$. The smallest positive integer is 3. Therefore, $\boxed{o(2) = 3}$.

Example 2.14. In $(\mathbb{Z}, +)$. The identity is 0. For $a = 3$, $3 + 3 + 3 + \cdots + 3 \neq 0$ for every positive integer number of additions. Hence, $\boxed{o(3) = \infty}$.

Note. In any group G with an identity element e. Since $e^1 = e$, the identity element always has order $\boxed{o(e) = 1}$.

The order of an element is the *number of times we need to combine the element with itself to get the identity for the first time.*

Definition 2.5 (Cyclic Group). For $a \in G$, write $\langle a \rangle = \{a^k : k \in \mathbb{Z}\}$.
A group G is **cyclic** if $G = \langle a \rangle$ for some $a \in G$, such an element a is called a **generator** of G.
- In additive notation, $\langle a \rangle = \{ka : k \in \mathbb{Z}\}$.

The powers $\ldots, a^{-3}, a^{-2}, a^{-1}, a^0 = e, a, a^2, a^3, \ldots$ need not be distinct.

Do the integer powers of an element form a subgroup?

Lemma 2.1. *For every $a \in G$, $\langle a \rangle$ is a subgroup of G.*

Proof

Let $a \in G$. Clearly, $\langle a \rangle \subseteq G$ by closure axiom and $\langle a \rangle \neq \varnothing$, since $a^0 = e \in \langle a \rangle$. Choose arbitrary $x, y \in \langle a \rangle$. By the definition of $\langle a \rangle$, there exist integers r, s such that $x = a^r$ and $y = a^s$. Therefore

$$\begin{aligned} xy^{-1} &= a^r(a^s)^{-1} && (\because \text{ substitution}) \\ &= a^r a^{-s} && (\because \text{ inverse law for powers}) \\ &= a^{r-s} && (\because \text{ product law for powers}) \\ &\in \langle a \rangle && (\because\ r - s \in \mathbb{Z}.) \end{aligned}$$

Since x and y were arbitrary, the one-step subgroup test shows that $\langle a \rangle$ is a subgroup of G. □

Is every cyclic group abelian?

Theorem 2.7. *Every cyclic group is abelian.*

Proof

Let $G = \langle a \rangle$ and choose arbitrary $x, y \in G$. By the definition of a cyclic group, there exist integers r, s such that $x = a^r$ and $y = a^s$. Therefore

$$\begin{aligned} xy &= a^r a^s && (\because \text{ substitution}) \\ &= a^{r+s} && (\because \text{ product law for powers}) \\ &= a^{s+r} && (\because \text{ commutativity of integer addition}) \\ &= a^s a^r && (\because \text{ product law for power})) \\ &= yx && (\because \text{ substitution.}) \end{aligned}$$

Since x and y were arbitrary, G is abelian. □

Does the inverse of a generator also generate the group?

Theorem 2.8. *If $G = \langle a \rangle$, then $G = \langle a^{-1} \rangle$. That is, if a is a generator of a cyclic group G, then a^{-1} is also a generator of G.*

Proof

Let $G = \langle a \rangle$ and let $x \in G$. Then $x = a^r$ for some integer r. Hence

$$x = a^r = (a^{-1})^{-r}.$$

Since $-r$ is an integer, x is an integer power of a^{-1}. Thus every element of G can be expressed as an integer power of a^{-1}. Therefore, a^{-1} also generates G. □

Is every subgroup of a cyclic group cyclic?

Theorem 2.9. *Every subgroup of a cyclic group is cyclic.*
(KUD: Jan-2026)(Question 36-QB)

Proof

Let $G = \langle a \rangle$ be a cyclic group and let H be a subgroup of G.
If $H = \{e\}$, then $H = \langle e \rangle$, so H is cyclic. Suppose $H \neq \{e\}$. Then H contains an element $a^t \neq e$, where t is a nonzero integer. If $t < 0$, then $a^{-t} = (a^t)^{-1} \in H$. Thus H contains a positive power of a.
Let m be the least positive integer such that $a^m \in H$. We claim that $H = \langle a^m \rangle$.
Let $x \in H$. Since $H \subseteq G = \langle a \rangle$, we have $x = a^t$ for some integer t. By the division algorithm, $t = qm + r$, with $0 \leq r < m$, for some integers q, r. Now

$$\begin{aligned} a^r &= a^{t-qm} && (\because \text{ division algorithm}) \\ &= a^t a^{-qm} && (\because \text{ product law for powers}) \\ &= x(a^m)^{-q} && (\because \text{ substitution and power-of-a-power law}) \\ &\in H && (\because \ x, a^m \in H \text{ and } H \text{ is a subgroup.}) \end{aligned}$$

If $r > 0$, then $a^r \in H$ with $r < m$, contradicting the choice of m as least positive integer. Therefore $r = 0$, and hence $x = a^t = a^{qm} = (a^m)^q \in \langle a^m \rangle$.
Thus every element $x \in H$ is an integral power of a^m.
Hence $H = \langle a^m \rangle$, so H is cyclic. □

When does a power of a generator also generate the group?

Theorem 2.10. *Let G be a cyclic group of order n generated by a. Then a^m generates G if and only if m and n are relatively prime.*
(Question 37-QB)

Proof

Since $G = \langle a \rangle$ has order n, the element a has order n. Thus $a^n = e$, where n is the least positive integer.
Suppose that $\gcd(m, n) = 1$, then there exist integers u, v such that $mu + nv = 1$.
Therefore

$$\begin{aligned} a &= a^{mu+nv} && (\because \ mu + nv = 1) \\ &= a^{mu} a^{nv} && (\because \text{ product law for powers}) \\ &= (a^m)^u (a^n)^v && (\because \text{ power-of-a-power law}) \\ &= (a^m)^u e && (\because \ a^n = e) \\ &= (a^m)^u && (\because \text{ identity axiom.}) \end{aligned}$$

Thus a is an integer power of a^m. Consequently, every integer power of a is also an integer power of a^m. Since $G = \langle a \rangle$, it follows that $G = \langle a^m \rangle$.

Conversely, suppose that $G = \langle a^m \rangle$. Let $d = \gcd(m, n)$. Since $a \in G$, there exists an integer k such that $a = (a^m)^k$.
Suppose $d > 1$. Then

$$\begin{aligned} a^{n/d} &= \left((a^m)^k\right)^{n/d} && (\because\ a = (a^m)^k) \\ &= a^{mkn/d} && (\because\ \text{power-of-a-power law}) \\ &= (a^n)^{(m/d)k} && (\because\ m/d \in \mathbb{Z}) \\ &= e && (\because\ a^n = e.) \end{aligned}$$

But $0 < n/d < n$, contradicting the fact that a has order n. Therefore $d = 1$, so m and n are relatively prime. □

Can powers of a generator repeat and how many generators does an infinite cyclic group have?

Theorem 2.11. *Let $G = \langle a \rangle$ be an infinite cyclic group. Then*
(i) Distinct integer powers of a are distinct; that is, $a^r = a^s$ implies $r = s$.
(ii) G has exactly two generators, namely a and a^{-1}.

(Question 38-QB)

Proof

Since $G = \langle a \rangle$ is an infinite cyclic group, no positive integer n can satisfy $a^n = e$.
(i) Suppose $a^r = a^s$ for integers r, s. If $r \neq s$, we may assume $r > s$. Then

$$\begin{aligned} a^{r-s} &= a^r a^{-s} && (\because\ \text{product law for powers}) \\ &= a^s a^{-s} && (\because\ a^r = a^s) \\ &= e && (\because\ \text{inverse law for powers and inverse axiom.}) \end{aligned}$$

Since $r - s > 0$, this contradicts the preceding observation. Hence $r = s$.
(ii) By assumption, a generates G. Also, for every integer r,

$$a^r = (a^{-1})^{-r}.$$

Thus every element of G is an integer power of a^{-1}, so a^{-1} also generates G.
Now let b be any generator of G. Since $b \in G = \langle a \rangle$, we have $b = a^m$ for some integer m. Since $a \in G = \langle b \rangle$, we have $a = b^k$ for some integer k. Therefore

$$\begin{aligned} a^1 &= b^k && (\because\ a = b^k) \\ &= (a^m)^k && (\because\ b = a^m) \\ &= a^{mk} && (\because\ \text{power-of-a-power law.}) \end{aligned}$$

By part (i), $mk = 1$. Since $m, k \in \mathbb{Z}$, either $m = k = 1$ or $m = k = -1$. Hence

$$b = a \qquad \text{or} \qquad b = a^{-1}.$$

Moreover, $a \neq a^{-1}$ by part (i), since $1 \neq -1$. Therefore G has exactly two generators, a and a^{-1}. □

Cosets and Lagrange's Theorem

Definition 2.6 (Left and right cosets). Let H be a subgroup of a group G, and let $a \in G$. Then

- the set $Ha = \{ha : h \in H\}$ is called the **right coset** of H in G determined by a.
- the set $aH = \{ah : h \in H\}$ is called the **left coset** of H in G determined by a.

☞ The element a is called a **representative** of the corresponding coset.

(Question 14-QB)

Basic observations. Let H be a subgroup of a group G with identity e, and let $a \in G$.

(i) Since $H \subseteq G$ and G is closed under multiplication, both Ha and aH are subsets of G.

(ii) Since $e \in H$,
$$a = ea \in Ha \qquad \text{and} \qquad a = ae \in aH.$$
Thus every coset contains its representative, and no coset is empty.

(iii) By the identity axiom,
$$He = H = eH.$$
Therefore H itself is both a right and a left coset.

(iv) If G is abelian, then $ah = ha$ for every $h \in H$. Hence $aH = Ha$.

If G is not abelian, the corresponding left and right cosets may or may not be equal.

☞ **In additive notation.** Let H be a subgroup of a group $(G, +)$, and let $a \in G$. Then the right and left cosets are respectively written as
$$H + a = \{h + a : h \in H\} \quad \text{and} \quad a + H = \{a + h : h \in H\}.$$

Example 2.15. Consider the multiplicative group of fourth roots of unity $G = \{1, -1, i, -i\}$ and its subgroup $H = \{1, -1\}$.
The right cosets of H in G are
$$\begin{aligned} H1 &= \{1 \cdot 1, (-1) \cdot 1\} = \{1, -1\} = H, \\ H(-1) &= \{1 \cdot (-1), (-1)(-1)\} = \{-1, 1\} = H, \\ Hi &= \{1 \cdot i, (-1)i\} = \{i, -i\}, \\ H(-i) &= \{1 \cdot (-i), (-1)(-i)\} = \{-i, i\} = Hi. \end{aligned}$$

Thus there are exactly two distinct right cosets:
$$H = \{1, -1\} \qquad \text{and} \qquad Hi = \{i, -i\}.$$

These cosets are disjoint and
$$G = H \cup Hi.$$

Since complex multiplication is commutative, the corresponding left and right cosets are equal.

Example 2.16 (Cosets of $3\mathbb{Z}$ in $\mathbb{Z}$)**.** Consider the additive group $(\mathbb{Z}, +)$ and its subgroup

$$H = 3\mathbb{Z} = \{\ldots, -6, -3, 0, 3, 6, \ldots\}.$$

Thus H consists of all multiples of 3.
To obtain the coset $H + a$, we add a to every element of H. Hence

$$\begin{aligned} H + 0 &= \{\ldots, -6 + 0, -3 + 0, 0 + 0, 3 + 0, 6 + 0, \ldots\} \\ &= \{\ldots, -6, -3, 0, 3, 6, \ldots\} = H, \\ H + 1 &= \{\ldots, -6 + 1, -3 + 1, 0 + 1, 3 + 1, 6 + 1, \ldots\} \\ &= \{\ldots, -5, -2, 1, 4, 7, \ldots\}, \\ H + 2 &= \{\ldots, -6 + 2, -3 + 2, 0 + 2, 3 + 2, 6 + 2, \ldots\} \\ &= \{\ldots, -4, -1, 2, 5, 8, \ldots\}. \end{aligned}$$

Adding 3, 4 or 5 produces no new cosets:

$$\begin{aligned} H + 3 &= \{\ldots, -6 + 3, -3 + 3, 0 + 3, 3 + 3, 6 + 3, \ldots\} \\ &= \{\ldots, -3, 0, 3, 6, 9, \ldots\} = H, \\ H + 4 &= \{\ldots, -6 + 4, -3 + 4, 0 + 4, 3 + 4, 6 + 4, \ldots\} \\ &= \{\ldots, -2, 1, 4, 7, 10, \ldots\} = H + 1, \\ H + 5 &= \{\ldots, -6 + 5, -3 + 5, 0 + 5, 3 + 5, 6 + 5, \ldots\} \\ &= \{\ldots, -1, 2, 5, 8, 11, \ldots\} = H + 2. \end{aligned}$$

Every integer has remainder 0, 1 or 2 when divided by 3. Accordingly,
(i) H contains all integers with remainder 0;
(ii) $H + 1$ contains all integers with remainder 1;
(iii) $H + 2$ contains all integers with remainder 2.
If a has remainder r, adding a to all multiples of 3 gives exactly the integers with remainder r. Hence $H + a = H + r$.
Therefore there are exactly three distinct right cosets:

$$H, \qquad H + 1, \qquad H + 2.$$

These cosets are pairwise disjoint and together contain every integer. Thus

$$\mathbb{Z} = H \cup (H + 1) \cup (H + 2).$$

Since addition is commutative, $a + H = H + a$. Therefore the corresponding left and right cosets are equal.

When does a coset equal the subgroup itself?

Lemma 2.2. *If H is a subgroup of a group G and $h \in H$, then $Hh = H = hH$.*

Proof

Let H be a subgroup of G, and let $h \in H$.
First, we prove that $Hh = H$. If $x \in Hh$, then $x = kh$ for some $k \in H$. Since H is a subgroup and $k, h \in H$, closure gives $x \in H$. Thus $Hh \subseteq H$.
Conversely, let $x \in H$. Then

$$\begin{aligned} x &= xe && (\because \text{identity axiom}) \\ &= x(h^{-1}h) && (\because \text{inverse axiom}) \\ &= (xh^{-1})h && (\because \text{associativity}) \\ &\in Hh && (\because xh^{-1} \in H, \text{ since } H \text{ is a subgroup}). \end{aligned}$$

Therefore $H \subseteq Hh$, and hence $Hh = H$.
Next, we prove that $hH = H$. Similar to the above, closure in the subgroup H gives $hH \subseteq H$. For any $x \in H$,

$$\begin{aligned} x &= ex && (\because \text{identity axiom}) \\ &= (hh^{-1})x && (\because \text{inverse axiom}) \\ &= h(h^{-1}x) && (\because \text{associativity}) \\ &\in hH && (\because h^{-1}x \in H, \text{ since } H \text{ is a subgroup}). \end{aligned}$$

Thus $H \subseteq hH$, so $hH = H$. □

Consequence. If H is a subgroup of a group G and $a \in G$, then

$$Ha = H \iff a \in H, \qquad aH = H \iff a \in H.$$

Indeed, if $Ha = H$, then $a \in Ha = H$. The converse follows from Lemma 2.2. The argument for left cosets is similar.

Notation used below. If H is a subgroup of a group G and $a, b \in G$, then $(Ha)b = H(ab)$, because $(ha)b = h(ab)$ for every $h \in H$. Similarly, $a(bH) = (ab)H$. These equalities follow from associativity in G.

When are two right cosets equal?

Theorem 2.12. *If H is a subgroup of a group G and $a, b \in G$, then*

$$Ha = Hb \iff ab^{-1} \in H.$$

(Question 39-QB)

Proof

Let H be a subgroup of G, and let $a, b \in G$.

Suppose $Ha = Hb$. Since $a \in Ha = Hb$. Thus $a = hb$ for some $h \in H$. Therefore

$$\begin{aligned} ab^{-1} &= (hb)b^{-1} && (\because a = hb) \\ &= h(bb^{-1}) && (\because \text{associativity}) \\ &= he && (\because \text{inverse axiom}) \\ &= h \in H && (\because \text{identity axiom}). \end{aligned}$$

Conversely, suppose $ab^{-1} \in H$. Since H is a subgroup, we have

$$H(ab^{-1}) = H.$$

Multiplying both sets on the right by b, we obtain

$$\Big(H(ab^{-1})\Big)b = Hb.$$

Now

$$\begin{aligned} \Big(H(ab^{-1})\Big)b &= H\Big((ab^{-1})b\Big) && (\because \text{associativity}) \\ &= H\Big(a(b^{-1}b)\Big) && (\because \text{associativity}) \\ &= H(ae) && (\because \text{inverse axiom}) \\ &= Ha && (\because \text{identity axiom}). \end{aligned}$$

Hence $Ha = Hb$. □

When are two left cosets equal?

Theorem 2.13. *If H is a subgroup of a group G and $a, b \in G$, then $aH = bH \iff a^{-1}b \in H$.*

Proof

Try this on your own. □

Can two right cosets overlap without being equal?

Theorem 2.14. *If H is a subgroup of a group G, then any two right cosets of H in G are either identical or disjoint.*

(Question 40-QB)

Proof

Let H be a subgroup of G, and let Ha and Hb be two right cosets of H.
If $Ha \cap Hb = \varnothing$, they are disjoint.

Suppose $Ha \cap Hb \neq \varnothing$. Choose $c \in Ha \cap Hb$. Then

$$c = h_1 a = h_2 b \text{ for some } h_1, h_2 \in H. \text{ Hence } \implies a = (h_1^{-1} h_2) b$$

Since H is a subgroup, $h_1^{-1} h_2 \in H$. Put $h_3 = h_1^{-1} h_2$. Then $a = h_3 b$, and

$$\begin{aligned} Ha &= H(h_3 b) && (\because a = h_3 b) \\ &= (Hh_3)b && (\because \text{associativity}) \\ &= Hb && (\because h_3 \in H, \text{ so } Hh_3 = H). \end{aligned}$$

Thus the cosets are identical whenever they intersect.
Therefore either $Ha = Hb$ or $Ha \cap Hb = \varnothing$. □

Theorem 2.15. *If H is a subgroup of a group G, then any two left cosets of H in G are either identical or disjoint.*

Proof

Try this on your own. □

How do cosets divide the group?

Theorem 2.16. *If H is a subgroup of a group G, then G is the union of all right cosets of H:*

$$G = \bigcup_{a \in G} Ha.$$

Proof

Let H be a subgroup of G. Since every right coset Ha is a subset of G,

$$\bigcup_{a \in G} Ha \subseteq G.$$

Conversely, let $x \in G$. Since $x \in Hx$, so x belongs to the union of all right cosets of H. Hence

$$G \subseteq \bigcup_{a \in G} Ha.$$

The two inclusions prove the result. □

Theorem 2.17. *If H is a subgroup of a group G, then G is the union of all left cosets of H:*

$$G = \bigcup_{a \in G} aH.$$

Proof

Try this on your own. □

Do all right cosets have the same number of elements?

Theorem 2.18. *If H is a subgroup of G, then prove that all right cosets of H in G have the same number of elements.*

(Question 41-QB)

Proof

Let H be a subgroup of G, and let Ha and Hb be two right cosets of H. Define

$$f : Ha \longrightarrow Hb, \qquad f(ha) = hb, \quad h \in H.$$

The function is well-defined. Every element of Ha has the form ha. If $h_1a = h_2a$, then $h_1 = h_2$ by right cancellation. Hence $h_1b = h_2b$. Thus the rule assigns a unique element of Hb to each element of Ha.
The function is one-to-one. Suppose $f(h_1a) = f(h_2a)$. Then

$$\begin{aligned} h_1b &= h_2b && (\because \text{definition of } f)\\ h_1 &= h_2 && (\because \text{right cancellation})\\ h_1a &= h_2a && (\because \text{multiplication on the right by } a). \end{aligned}$$

Therefore f is one-to-one.
The function is onto. Let $y \in Hb$. Then $y = hb$ for some $h \in H$. The element ha belongs to Ha, and

$$f(ha) = hb = y.$$

Hence f is onto.
Thus f is a bijection, so $|Ha| = |Hb|$. Since $H = He$ is itself a right coset, every right coset has the same cardinality as H. □

Theorem 2.19. *If H is a subgroup of G, then prove that all left cosets of H in G have the same number of elements.*

Proof

Try this on your own. □

How is the order of a subgroup related to the order of the group?

Lagrange's theorem

Theorem 2.20 (Lagrange's theorem). *The order of every subgroup of a finite group divides the order of the group.*

(KUD: Jan-2026)(Question 42-QB)

Proof

Let H be a subgroup of a finite group G, with $|G| = n$ and $|H| = m$.
Since G is finite, there are finitely many distinct right cosets of H in G.
Let them be $Ha_1, Ha_2, \ldots, Ha_k$.
By the right coset decomposition, $G = Ha_1 \cup Ha_2 \cup \cdots \cup Ha_k$, where these cosets are pairwise disjoint.
But, each coset contains exactly m elements. Therefore

$$\begin{aligned} n = |G| &= |Ha_1| + |Ha_2| + \cdots + |Ha_k| && (\because \text{the cosets are pairwise disjoint}) \\ &= \underbrace{m + m + \cdots + m}_{k \text{ terms}} && (\because \text{each coset has } m \text{ elements}) \\ &= km. \end{aligned}$$

Since k is a positive integer, m divides n. Hence $|H|$ divides $|G|$. □

Permutations

Definition 2.7. A **permutation** of a nonempty set S is a bijective function from S onto itself.

☞ Thus a permutation rearranges the elements of S so that every element appears exactly once as an image.

Two-row notation. A permutation σ of $S = \{a, b, c\}$ is written as

$$\sigma = \begin{pmatrix} a & b & c \\ \sigma(a) & \sigma(b) & \sigma(c) \end{pmatrix}.$$

The first row lists the elements of S, and the second row lists their respective images.

Example 2.17. Let $S = \{a, b\}$. The function $\sigma = \begin{pmatrix} a & b \\ b & a \end{pmatrix}$ interchanges a and b: $\sigma(a) = b$ and $\sigma(b) = a$. The images are distinct and include every element of S. Hence σ is bijective and is therefore a permutation of S.

Example 2.18. Let $S = \{a, b, c\}$. The function $\tau = \begin{pmatrix} a & b & c \\ b & c & a \end{pmatrix}$ sends a to b, b to c, and c to a. Its images are distinct and include every element of S. Hence τ is bijective and is therefore a permutation of S.

How do we compose two permutations?

Definition 2.8 (Composition of permutations). Let σ and τ be permutations of a set S. Their composition $\sigma \circ \tau$ is defined by $(\sigma \circ \tau)(x) = \sigma(\tau(x))$ for every $x \in S$. Thus we apply τ first and then σ.

Convention. Throughout this section, $\circ$ denotes ordinary function composition, evaluated from right to left.

Convention. We use ordinary function composition: in $\sigma \circ \tau$, apply τ first and then σ. A cycle (123) sends $1 \mapsto 2$, $2 \mapsto 3$ and $3 \mapsto 1$. A symbol not appearing in a cycle is fixed. For example, $(123)(4) = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 1 & 4 \end{pmatrix}$.

Example 2.19. Let $\sigma = \begin{pmatrix} a & b & c \\ a & c & b \end{pmatrix}$ and $\tau = \begin{pmatrix} a & b & c \\ c & b & a \end{pmatrix}$. Find $\sigma \circ \tau$.

Solution

By the definition of composition,

$$(\sigma \circ \tau)(a) = \sigma(\tau(a)) = \sigma(c) = b,$$
$$(\sigma \circ \tau)(b) = \sigma(\tau(b)) = \sigma(b) = c,$$
$$(\sigma \circ \tau)(c) = \sigma(\tau(c)) = \sigma(a) = a.$$

Therefore

$$\sigma \circ \tau = \begin{pmatrix} a & b & c \\ b & c & a \end{pmatrix}.$$

Example 2.20. Let $\sigma = \begin{pmatrix} a & b & c \\ b & c & a \end{pmatrix}$ and $\tau = \begin{pmatrix} a & b & c \\ c & a & b \end{pmatrix}$. Find $\sigma \circ \tau$.

Solution

By the definition of composition,

$$(\sigma \circ \tau)(a) = \sigma(\tau(a)) = \sigma(c) = a,$$
$$(\sigma \circ \tau)(b) = \sigma(\tau(b)) = \sigma(a) = b,$$
$$(\sigma \circ \tau)(c) = \sigma(\tau(c)) = \sigma(b) = c.$$

Therefore

$$\sigma \circ \tau = \begin{pmatrix} a & b & c \\ a & b & c \end{pmatrix} = e.$$

Do the permutations of $\{a, b, c\}$ form a group?

Problem 2.16. Prove that all permutations of the set $S = \{a, b, c\}$ form a group under composition of permutations.

(Question 35-QB)

Solution

Let G be the set of all permutations of $S = \{a, b, c\}$. There are $3! = 6$ such permutations:

$$e = \begin{pmatrix} a & b & c \\ a & b & c \end{pmatrix}, \quad \alpha = \begin{pmatrix} a & b & c \\ b & a & c \end{pmatrix}, \quad \beta = \begin{pmatrix} a & b & c \\ a & c & b \end{pmatrix},$$

$$\gamma = \begin{pmatrix} a & b & c \\ c & b & a \end{pmatrix}, \quad \rho = \begin{pmatrix} a & b & c \\ b & c & a \end{pmatrix}, \quad \lambda = \begin{pmatrix} a & b & c \\ c & a & b \end{pmatrix}.$$

Thus

$$G = \{e, \alpha, \beta, \gamma, \rho, \lambda\}.$$

We verify the group axioms.

1. **Closure.**
   Let $\sigma, \tau \in G$. Since both are functions from S to S, their composition is also a function from S to S.
   We show that $\sigma \circ \tau$ is bijective.
   One-to-one. Let $x, y \in S$ and suppose $(\sigma \circ \tau)(x) = (\sigma \circ \tau)(y)$. Then

   $$\begin{aligned} \sigma(\tau(x)) &= \sigma(\tau(y)) && (\because \text{ definition of composition}) \\ \tau(x) &= \tau(y) && (\because \sigma \text{ is one-to-one}) \\ x &= y && (\because \tau \text{ is one-to-one}). \end{aligned}$$

   Hence $\sigma \circ \tau$ is one-to-one.
   Onto. Let $z \in S$. Since σ is onto, there exists $y \in S$ such that $\sigma(y) = z$. Since τ is onto, there exists $x \in S$ such that $\tau(x) = y$. Therefore

   $$(\sigma \circ \tau)(x) = \sigma(\tau(x)) = \sigma(y) = z.$$

   Hence $\sigma \circ \tau$ is onto.
   Thus $\sigma \circ \tau$ is a permutation of S, so $\sigma \circ \tau \in G$.
2. **Associativity.**
   Let $\sigma, \tau, \nu \in G$. For every $x \in S$,

   $$\begin{aligned} ((\sigma \circ \tau) \circ \nu)(x) &= (\sigma \circ \tau)(\nu(x)) && (\because \text{ definition of composition}) \\ &= \sigma(\tau(\nu(x))) && (\because \text{ definition of composition}) \\ &= \sigma((\tau \circ \nu)(x)) && (\because \text{ definition of composition}) \\ &= (\sigma \circ (\tau \circ \nu))(x) && (\because \text{ definition of composition}). \end{aligned}$$

   Since the two functions agree at every $x \in S$,

   $$(\sigma \circ \tau) \circ \nu = \sigma \circ (\tau \circ \nu).$$

   Therefore composition is associative.

3. **Identity.**
   The identity permutation $e \in G$ is defined by $e(x) = x$ for every $x \in S$. For any $\sigma \in G$ and $x \in S$,
   $$(e \circ \sigma)(x) = e(\sigma(x)) = \sigma(x),$$
   and
   $$(\sigma \circ e)(x) = \sigma(e(x)) = \sigma(x).$$
   Hence
   $$e \circ \sigma = \sigma = \sigma \circ e.$$
   Thus e is the identity element.
4. **Inverses.**
   Observe the following:
   $$e \circ e = e, \quad \alpha \circ \alpha = e, \quad \beta \circ \beta = e, \quad \gamma \circ \gamma = e, \quad \text{and} \quad \rho \circ \lambda = e = \lambda \circ \rho.$$
   Hence the inverses are
   $$e^{-1} = e, \quad \alpha^{-1} = \alpha, \quad \beta^{-1} = \beta, \quad \gamma^{-1} = \gamma, \quad \rho^{-1} = \lambda, \quad \lambda^{-1} = \rho.$$
   Therefore every element of G has an inverse in G.

Hence $(G, \circ)$ is a group of order 6.

☞ The above group is called the **symmetric group on three elements** and is denoted by S_3.

Normal Subgroups

For a subgroup H of a group G, the left coset aH and the right coset Ha need not be equal. Subgroups for which these cosets are equal for every $a \in G$ are called *normal subgroups*.

Definition 2.9 (Normal subgroup)**.** A subgroup H of a group G is called a **normal subgroup** of G if $aH = Ha$ for every $a \in G$.

(Question 15-QB)

Note. Every subgroup of an abelian group is normal. Indeed, if G is abelian and H is a subgroup of G, then $aH = \{ah : h \in H\} = \{ha : h \in H\} = Ha$ for every $a \in G$.

Example 2.21. Consider the multiplicative group of fourth roots of unity $G = \{1, -1, i, -i\}$ and its subgroup $H = \{1, -1\}$. We have
$$\begin{aligned} 1H &= H1 = \{1, -1\}, \\ (-1)H &= H(-1) = \{1, -1\}, \\ iH &= Hi = \{i, -i\}, \\ (-i)H &= H(-i) = \{i, -i\}. \end{aligned}$$

Thus $aH = Ha$ for every $a \in G$. Therefore H is a normal subgroup of G.

Example 2.22. Consider the permutation group

$$G = S_3 = \{e, (ab)(c), (bc)(a), (ac)(b), (abc), (acb)\},$$

where $e = (a)(b)(c)$, and its subgroup $H = \{e, (ab)(c)\}$.
We use ordinary function composition: in $\sigma \circ \tau$, we apply τ first and then σ.
Let $\sigma = (ab)(c)$ and $g = (bc)(a)$. Then

$$\sigma = \begin{pmatrix} a & b & c \\ b & a & c \end{pmatrix} \text{ and } g = \begin{pmatrix} a & b & c \\ a & c & b \end{pmatrix}.$$

$\sigma \circ g$ is

$$\sigma \circ g = \begin{pmatrix} a & b & c \\ b & c & a \end{pmatrix} = (abc).$$

Therefore the right coset is

$$Hg = \{e \circ g, \sigma \circ g\} = \{(bc)(a), (abc)\}.$$

$g \circ \sigma$ is

$$g \circ \sigma = \begin{pmatrix} a & b & c \\ c & a & b \end{pmatrix} = (acb).$$

Therefore the left coset is

$$gH = \{g \circ e, g \circ \sigma\} = \{(bc)(a), (acb)\}.$$

Since $Hg \neq gH$, the subgroup H is not normal in G.

How can we test whether a subgroup is normal?

Conjugation. For $a, h \in G$, the element $a^{-1}ha$ is called a conjugate of h. If H is a subgroup of G, we write $a^{-1}Ha = \{a^{-1}ha : h \in H\}$.

Theorem 2.21. *A subgroup H of a group G is normal if and only if $a^{-1}ha \in H$ for all $h \in H$ and $a \in G$.*

(Question 43-QB)

Proof

Let H be a subgroup of G.

Suppose H is normal in G. Let $h \in H$ and $a \in G$. Since H is normal, $Ha = aH$. Now $ha \in Ha = aH$, so $ha = ak$ for some $k \in H$. Therefore

$$\begin{aligned} a^{-1}ha &= a^{-1}(ha) && (\because \text{associativity}) \\ &= a^{-1}(ak) && (\because ha = ak) \\ &= (a^{-1}a)k && (\because \text{associativity}) \\ &= ek && (\because \text{inverse axiom}) \\ &= k \in H && (\because \text{identity axiom}). \end{aligned}$$

Conversely, suppose $a^{-1}ha \in H$ for all $h \in H$ and $a \in G$. We show that $Ha = aH$ for every $a \in G$.
Fix $a \in G$. Let $x \in Ha$. Then $x = ha$ for some $h \in H$, and

$$\begin{aligned} x &= ha && (\because \text{substitution}) \\ &= (aa^{-1})ha && (\because \text{inverse and identity axioms}) \\ &= a(a^{-1}ha) && (\because \text{associativity}) \\ &\in aH && (\because a^{-1}ha \in H \text{ by assumption}). \end{aligned}$$

Thus $Ha \subseteq aH$.
To prove the reverse inclusion, observe that $a^{-1} \in G$. Applying the assumption with a^{-1} in place of a gives $(a^{-1})^{-1}ha^{-1} = aha^{-1} \in H$ for every $h \in H$.
Let $y \in aH$. Then $y = ah$ for some $h \in H$, and

$$\begin{aligned} y &= ah && (\because \text{substitution}) \\ &= ah(a^{-1}a) && (\because \text{inverse and identity axioms}) \\ &= (aha^{-1})a && (\because \text{associativity}) \\ &\in Ha && (\because aha^{-1} \in H). \end{aligned}$$

Thus $aH \subseteq Ha$.
Hence $aH = Ha$ for every $a \in G$. Therefore H is normal in G. □

Does conjugating a subgroup produce another subgroup?

Theorem 2.22. *If H is a subgroup of a group G and $g \in G$, then $g^{-1}Hg = \{g^{-1}hg : h \in H\}$ is also a subgroup of G.*

(Question 44-QB)

Proof

Let H be a subgroup of G, and fix $g \in G$. Put $K = g^{-1}Hg$. Clearly, $K \subseteq G$, by closure in G.
Since H is a subgroup, $e \in H$. Hence $g^{-1}eg = g^{-1}g = e \in K$. Thus $K \neq \varnothing$.
Let $x, y \in K$. Then $x = g^{-1}h_1g, y = g^{-1}h_2g$ for some $h_1, h_2 \in H$.

First,

$$\begin{aligned} y^{-1} &= (g^{-1}h_2g)^{-1} && (\because \text{substitution}) \\ &= g^{-1}h_2^{-1}(g^{-1})^{-1} && (\because \text{inverse of a product}) \\ &= g^{-1}h_2^{-1}g && (\because (g^{-1})^{-1} = g). \end{aligned}$$

Therefore

$$\begin{aligned} xy^{-1} &= (g^{-1}h_1g)(g^{-1}h_2^{-1}g) && (\because \text{substitution}) \\ &= g^{-1}h_1(gg^{-1})h_2^{-1}g && (\because \text{associativity}) \\ &= g^{-1}h_1eh_2^{-1}g && (\because \text{inverse axiom}) \\ &= g^{-1}(h_1h_2^{-1})g && (\because \text{identity axiom and associativity}) \\ &\in K && (\because h_1h_2^{-1} \in H, \text{ since } H \text{ is a subgroup}). \end{aligned}$$

Thus $xy^{-1} \in K$ for every $x, y \in K$.
By the one-step subgroup test, $K = g^{-1}Hg$ is a subgroup of G. □

Remark. The subgroup $g^{-1}Hg$ is called a **conjugate subgroup** of H. It need not equal H. A subgroup H is normal precisely when $g^{-1}Hg = H$ for every $g \in G$.

Quotient Groups

A normal subgroup allows us to treat its cosets as the elements of a new group. The operation on these cosets is obtained from the operation of the original group.

Definition 2.10 (Quotient group). Let H be a normal subgroup of a group G. The set of all left cosets of H in G is denoted by $G/H = \{xH : x \in G\}$.
Under the operation $(xH)\cdot(yH) = (xy)H$, this set forms a group called the **quotient group** or **factor group** of G by H.

(Question 16-QB)

- The elements of G/H are cosets, which are subsets of G. The identity element of G/H is the coset $H = eH$.
- For an additively written group, the operation is

$$(x + H) + (y + H) = (x + y) + H.$$

Example 2.23 (Integers modulo 2)**.** Consider the additive group $(\mathbb{Z}, +)$ and its subgroup $H = 2\mathbb{Z}$. Since $\mathbb{Z}$ is abelian, H is normal.
There are two distinct cosets:

$$H = \{\ldots, -4, -2, 0, 2, 4, \ldots\} \text{ and } 1 + H = \{\ldots, -3, -1, 1, 3, 5, \ldots\}.$$

Thus

$$\mathbb{Z}/2\mathbb{Z} = \{H, 1 + H\}.$$

Addition of cosets gives

$$\begin{aligned} H + H &= H, \\ H + (1 + H) &= 1 + H, \\ (1 + H) + H &= 1 + H, \\ (1 + H) + (1 + H) &= 2 + H = H. \end{aligned}$$

Here $2 + H = H$ because $2 \in H$.
The Cayley table is

$+$	H	$1+H$
H	H	$1+H$
$1+H$	$1+H$	H

Hence $\mathbb{Z}/2\mathbb{Z}$ is a quotient group of order 2, with identity H.

Example 2.24 (Fourth roots of unity)**.** Consider the multiplicative group $G = \{1, -1, i, -i\}$ and its subgroup $H = \{1, -1\}$. Since G is abelian, H is normal.
The distinct cosets are $H = \{1, -1\}$ and $iH = \{i, -i\}$.
Therefore

$$G/H = \{H, iH\}.$$

Multiplication of cosets gives

$$\begin{aligned} H \cdot H &= H, \\ H \cdot iH &= iH, \\ iH \cdot H &= iH, \\ iH \cdot iH &= i^2 H = (-1)H = H. \end{aligned}$$

The last equality holds because $-1 \in H$.
The Cayley table is

$\cdot$	H	iH
H	H	iH
iH	iH	H

Thus G/H is a quotient group of order 2. Its identity is H, and iH is its own inverse.

Do the set of cosets of a normal subgroup form a group?

Theorem 2.23. *Let H be a normal subgroup of a group G.*
Then the set $G/H = \{xH : x \in G\}$ forms a group under the operation $(xH) \cdot (yH) = (xy)H$.

(Question 45-QB)

Proof

Let H be a normal subgroup of G. Since $H = eH \in G/H$, the set G/H is nonempty. We now verify the group axioms.

1. **Closure.**
   Let $xH, yH \in G/H$. Since $x, y \in G$ and G is a group, $xy \in G$. Therefore
   $$(xH) \cdot (yH) = (xy)H \in G/H.$$

2. **Associativity.**
   Let $xH, yH, zH \in G/H$. Then
   $$\begin{aligned}
   \Big((xH) \cdot (yH)\Big) \cdot (zH) &= ((xy)H) \cdot (zH) && (\because \text{definition of the operation}) \\
   &= ((xy)z)H && (\because \text{definition of the operation}) \\
   &= (x(yz))H && (\because \text{associativity in } G) \\
   &= (xH) \cdot ((yz)H) && (\because \text{definition of the operation}) \\
   &= (xH) \cdot \Big((yH) \cdot (zH)\Big) && (\because \text{definition of the operation}).
   \end{aligned}$$
   Hence the operation is associative.

3. **Identity.**
   Since $eH = H$, for every $xH \in G/H$,
   $$\begin{aligned}
   H \cdot (xH) &= (eH) \cdot (xH) = (ex)H = xH, \\
   (xH) \cdot H &= (xH) \cdot (eH) = (xe)H = xH.
   \end{aligned}$$
   These equalities follow from the definition of the operation and the identity axiom in G. Thus H is the identity element of G/H.

4. **Inverses.**
   Let $xH \in G/H$. Since $x^{-1} \in G$, the coset $x^{-1}H$ belongs to G/H. Moreover,
   $$\begin{aligned}
   (xH) \cdot (x^{-1}H) &= (xx^{-1})H = eH = H, \\
   (x^{-1}H) \cdot (xH) &= (x^{-1}x)H = eH = H.
   \end{aligned}$$
   These equalities follow from the definition of the operation and the inverse axiom in G. Therefore
   $$(xH)^{-1} = x^{-1}H.$$

Hence G/H is a group under the given operation. □

Note. If H is a normal subgroup of a finite group G, then Lagrange's theorem gives
$$|G/H| = \frac{|G|}{|H|}.$$

A quotient group of a cyclic group of order 9

Problem 2.17. Let G be a cyclic group of order 9 generated by a, and H be a subgroup of G generated by a^3 then prepare the Cayley table for G/H.
(Question 46-QB)

Solution

Since $G = \langle a \rangle$ has order 9,

$$G = \{e, a, a^2, a^3, a^4, a^5, a^6, a^7, a^8\}, \qquad a^9 = e.$$

Determine the subgroup H:
The powers of a^3 are

$$(a^3)^0 = e, \qquad (a^3)^1 = a^3, \qquad (a^3)^2 = a^6, \qquad (a^3)^3 = a^9 = e.$$

Hence

$$H = \langle a^3 \rangle = \{e, a^3, a^6\}.$$

Since G is cyclic, it is abelian. Every subgroup of an abelian group is normal, so H is normal in G.
Find the distinct cosets:
Using the product law for powers,

$$\begin{aligned} H &= \{e, a^3, a^6\}, \\ aH &= \{ae, aa^3, aa^6\} = \{a, a^4, a^7\}, \\ a^2H &= \{a^2e, a^2a^3, a^2a^6\} = \{a^2, a^5, a^8\}. \end{aligned}$$

These three cosets are distinct and together contain every element of G. Therefore

$$G/H = \{H, aH, a^2H\}.$$

Also,

$$|G/H| = \frac{|G|}{|H|} = \frac{9}{3} = 3.$$

Compute the products of cosets:
The operation is

$$(a^rH) \cdot (a^sH) = a^{r+s}H.$$

Since $a^3 \in H$, we have $a^3H = H$. Consequently,

$$\begin{aligned} (aH) \cdot (aH) &= a^2H && (\because \text{definition of coset multiplication}) \\ (aH) \cdot (a^2H) &= a^3H = H && (\because a^3 \in H) \\ (a^2H) \cdot (aH) &= a^3H = H && (\because a^3 \in H) \\ (a^2H) \cdot (a^2H) &= a^4H = a(a^3H) = aH && (\because a^3H = H). \end{aligned}$$

Since H is the identity coset, the Cayley table is

$\cdot$	H	aH	a^2H
H	H	aH	a^2H
aH	aH	a^2H	H
a^2H	a^2H	H	aH

Homomorphisms and Isomorphisms of Groups

Definition 2.11. Let $(G, \cdot)$ and $(G', *)$ be two groups. A function $f : G \to G'$ is called a **group homomorphism** if $f(ab) = f(a) * f(b)$ for all $a, b \in G$.
Thus a homomorphism preserves the group operation.

(Question 17-QB)

☞ **Note on notation.** The homomorphism condition depends on the operations in the two groups:

Operation in G	Operation in G'	Homomorphism condition
Multiplication	Multiplication	$f(ab) = f(a)f(b)$
Addition	Addition	$f(a+b) = f(a) + f(b)$
Addition	Multiplication	$f(a+b) = f(a)f(b)$
Multiplication	Addition	$f(ab) = f(a) + f(b)$

Definition 2.12. A bijective homomorphism $f : G \to G'$ is called a **group isomorphism**. If such a map exists, the groups G and G' are called **isomorphic**, and we write $G \cong G'$.

(Question 18-QB)

☞ **How to prove that a map is an isomorphism.** Verify that the map:

(i) preserves the group operation;

(ii) is one-to-one;

(iii) is onto.

What happens to identities and inverses under a homomorphism?

Problem 2.18. Let $f : G \to G'$ be a homomorphism of groups, and let e and e' be the identity elements of G and G', respectively. Prove that:
(i) $f(e) = e'$;
(ii) $f(a^{-1}) = [f(a)]^{-1}$ for every $a \in G$.

(Question 47-QB)

Solution

(i). Since $ee = e$ in G,

$$\begin{aligned} f(e)f(e) &= f(ee) && (\because f \text{ is a homomorphism}) \\ &= f(e) && (\because ee = e) \\ &= f(e)e' && (\because e' \text{ is the identity in } G'). \end{aligned}$$

By the left cancellation law in G', $f(e) = e'$.

(ii). Let $a \in G$. Then

$$\begin{aligned} f(a)f(a^{-1}) &= f(aa^{-1}) && (\because f \text{ is a homomorphism}) \\ &= f(e) && (\because \text{inverse axiom in } G) \\ &= e' && (\because \text{part (i)}). \end{aligned}$$

Similarly,

$$\begin{aligned} f(a^{-1})f(a) &= f(a^{-1}a) && (\because f \text{ is a homomorphism}) \\ &= f(e) && (\because \text{inverse axiom in } G) \\ &= e' && (\because \text{part (i)}). \end{aligned}$$

Hence $f(a^{-1})$ is the inverse of $f(a)$ in G'. By uniqueness of inverses,

$$f(a^{-1}) = [f(a)]^{-1}.$$

☞ **Useful consequence:** If $f : G \to G'$ is a homomorphism, then $f(a^m) = [f(a)]^m$ for every $a \in G$ and every $m \in \mathbb{Z}$.

What happens to subgroups under a homomorphism?

Problem 2.19. Let $f : G \to G'$ be a homomorphism of groups. Prove that:
(i) If H is a subgroup of G, then $f(H)$ is a subgroup of G'.
(ii) If H is a cyclic subgroup of G, then $f(H)$ is a cyclic subgroup of G'.
(Question 48-QB)

Solution

(i) The image of a subgroup is a subgroup.
Let H be a subgroup of G. Its image is

$$f(H) = \{f(h) : h \in H\} \subseteq G'.$$

Since $e \in H$ and $f(e) = e'$, we have $e' \in f(H)$. Thus $f(H) \neq \varnothing$.
Let $x, y \in f(H)$. Then $x = f(h_1)$ and $y = f(h_2)$ for some $h_1, h_2 \in H$. Therefore

$$\begin{aligned} xy^{-1} &= f(h_1)[f(h_2)]^{-1} && (\because \text{substitution}) \\ &= f(h_1)f(h_2^{-1}) && (\because \text{preservation of inverses}) \\ &= f(h_1h_2^{-1}) && (\because f \text{ is a homomorphism}) \\ &\in f(H) && (\because h_1h_2^{-1} \in H, \text{ since } H \text{ is a subgroup}). \end{aligned}$$

By the one-step subgroup test, $f(H)$ is a subgroup of G'.
(ii) The image of a cyclic subgroup is cyclic.
Let $H = \langle a \rangle$ be a cyclic subgroup of G. By part (i), $f(H)$ is a subgroup of G'.

Every element of H is a^m for some integer m. Hence

$$\begin{aligned} f(H) &= \{f(a^m) : m \in \mathbb{Z}\} && (\because H = \langle a \rangle) \\ &= \{[f(a)]^m : m \in \mathbb{Z}\} && (\because \text{preservation of integer powers}) \\ &= \langle f(a) \rangle && (\because \text{definition of a cyclic subgroup}). \end{aligned}$$

Therefore $f(H)$ is cyclic, generated by $f(a)$.

Which familiar group represents every infinite cyclic group?

Problem 2.20. Prove that every infinite cyclic group is isomorphic to the additive group $(\mathbb{Z}, +)$.

(Question 49-QB)

Solution

Let $G = \langle a \rangle$ be an infinite cyclic group. Define $f : \mathbb{Z} \to G$ by $f(m) = a^m$.
Homomorphism. For all $m, n \in \mathbb{Z}$,

$$\begin{aligned} f(m+n) &= a^{m+n} && (\because \text{definition of } f) \\ &= a^m a^n && (\because \text{product law for powers}) \\ &= f(m)f(n) && (\because \text{definition of } f). \end{aligned}$$

Thus f is a homomorphism.
One-to-one. Suppose $f(m) = f(n)$. Then $a^m = a^n$. Distinct integer powers of the generator of an infinite cyclic group are distinct, so $m = n$. Hence f is one-to-one.
Onto. Let $x \in G$. Since $G = \langle a \rangle$, there exists an integer m such that $x = a^m$. Therefore $f(m) = x$, so f is onto.
Hence f is an isomorphism, and $G \cong (\mathbb{Z}, +)$.

Which familiar group represents every finite cyclic group?

Problem 2.21. Prove that every cyclic group of order n is isomorphic to the additive group $(\mathbb{Z}_n, +_n)$ of integers modulo n.

(Question 50-QB)

Solution

Let $G = \langle a \rangle$ be a cyclic group of order n. Then $G = \{e, a, a^2, \ldots, a^{n-1}\}$, where $a^n = e$, and a has order n.
Using the set $\mathbb{Z}_n = \{0, 1, \ldots, n-1\}$, define $f : \mathbb{Z}_n \to G$ such that $f(r) = a^r$.
Homomorphism. Let $r, s \in \mathbb{Z}_n$. By the division algorithm,

$$r + s = qn + t, \qquad 0 \le t < n.$$

Thus $r +_n s = t$. Also,

$$\begin{aligned} f(r)f(s) &= a^r a^s && (\because \text{definition of } f) \\ &= a^{r+s} && (\because \text{product law for powers}) \\ &= a^{qn+t} && (\because \text{division algorithm}) \\ &= (a^n)^q a^t && (\because \text{laws of powers}) \\ &= ea^t && (\because a^n = e) \\ &= a^t && (\because \text{identity axiom}) \\ &= f(r +_n s) && (\because r +_n s = t). \end{aligned}$$

Therefore f is a homomorphism.
One-to-one. Suppose $f(r) = f(s)$, where $0 \leq r, s < n$. Then $a^r = a^s$.
If $r \neq s$, we may assume $r > s$. Multiplication by a^{-s} gives

$$a^{r-s} = e.$$

But $0 < r - s < n$, contradicting the fact that a has order n. Hence $r = s$, and f is one-to-one.
Onto. Every element of G is a^r for some $r \in \{0, 1, \ldots, n-1\}$. Since $f(r) = a^r$, the map f is onto.
Therefore f is an isomorphism, and $G \cong (\mathbb{Z}_n, +_n)$.

Problem 2.22. Let $P = \mathbb{R}^+$ be the multiplicative group of positive real numbers. Define $f : (P, \cdot) \to (\mathbb{R}, +)$ by $f(x) = \log_{10} x$. Prove that f is a homomorphism and determine whether it is an isomorphism.

(Question 51-QB)

Solution

Since $\log_{10} x$ is a real number for every $x > 0$, the map is well-defined.
Homomorphism. For all $x, y \in P$,

$$\begin{aligned} f(xy) &= \log_{10}(xy) && (\because \text{definition of } f) \\ &= \log_{10} x + \log_{10} y && (\because \text{product law for logarithms}) \\ &= f(x) + f(y) && (\because \text{definition of } f). \end{aligned}$$

Thus f is a homomorphism.
One-to-one. Suppose $f(x) = f(y)$. Then

$$\log_{10} x = \log_{10} y.$$

Taking powers of 10 on both sides gives

$$x = 10^{\log_{10} x} = 10^{\log_{10} y} = y.$$

Hence f is one-to-one.

Onto. Let $r \in \mathbb{R}$. Choose $x = 10^r > 0$. Then

$$f(x) = \log_{10}(10^r) = r.$$

Hence f is onto.
Therefore f is an isomorphism, and $(\mathbb{R}^+, \cdot) \cong (\mathbb{R}, +)$.

Problem 2.23. Define $f : (\mathbb{R}, +) \to (\mathbb{R}^+, \cdot)$ by $f(x) = e^x$. Prove that f is an isomorphism.

(Question 52-QB)

Solution

Since $e^x > 0$ for every $x \in \mathbb{R}$, the map is well-defined.
Homomorphism. For all $x, y \in \mathbb{R}$,

$$\begin{aligned} f(x+y) &= e^{x+y} && (\because \text{definition of } f) \\ &= e^x e^y && (\because \text{law of exponents}) \\ &= f(x)f(y) && (\because \text{definition of } f). \end{aligned}$$

Therefore f is a homomorphism.
One-to-one. Suppose $f(x) = f(y)$. Then $e^x = e^y$. Taking natural logarithms gives

$$x = \ln(e^x) = \ln(e^y) = y.$$

Thus f is one-to-one.
Onto. Let $t \in \mathbb{R}^+$. Choose $x = \ln t \in \mathbb{R}$. Then

$$f(x) = e^{\ln t} = t.$$

Hence f is onto.
Therefore f is an isomorphism, and $(\mathbb{R}, +) \cong (\mathbb{R}^+, \cdot)$.

Problem 2.24. Define $g : (\mathbb{R}^+, \cdot) \to (\mathbb{R}, +)$ by $g(x) = \log x$. Prove that g is an isomorphism.

(Question 53-QB)

Solution

Since $\log x \in \mathbb{R}$ for every $x > 0$, the map is well-defined.
Homomorphism. For all $x, y \in \mathbb{R}^+$,

$$\begin{aligned} g(xy) &= \log(xy) && (\because \text{definition of } g) \\ &= \log x + \log y && (\because \text{product law for logarithms}) \\ &= g(x) + g(y) && (\because \text{definition of } g). \end{aligned}$$

Hence g is a homomorphism.

One-to-one. Suppose $g(x) = g(y)$. Then $\log x = \log y$. Applying the exponential function gives

$$x = e^{\log x} = e^{\log y} = y.$$

Thus g is one-to-one.
Onto. Let $r \in \mathbb{R}$. Choose $x = e^r > 0$. Then

$$g(x) = \log(e^r) = r.$$

Therefore g is onto.
Hence g is an isomorphism. It is the inverse of the exponential isomorphism in Question 52.

Are the integers and the even integers isomorphic?

Problem 2.25. Let $(\mathbb{Z}, +)$ be the additive group of integers, and let $(E, +)$ be the additive group of even integers, where $E = 2\mathbb{Z}$. Define $f : \mathbb{Z} \to E$ by $f(x) = 2x$. Prove that f is an isomorphism.

(Question 54-QB)

Solution

For every $x \in \mathbb{Z}$, the integer $2x$ is even. Hence $f(x) \in E$, so f is well-defined.
Homomorphism. For all $x, y \in \mathbb{Z}$,

$$\begin{aligned} f(x+y) &= 2(x+y) && (\because \text{definition of } f) \\ &= 2x + 2y && (\because \text{distributive law in } \mathbb{Z}) \\ &= f(x) + f(y) && (\because \text{definition of } f). \end{aligned}$$

Hence f is a homomorphism.
One-to-one. Suppose $f(x) = f(y)$. Then $2x = 2y$, so $x = y$. Therefore f is one-to-one.
Onto. Let $t \in E$. Since t is even, $t = 2k$ for some $k \in \mathbb{Z}$. Then

$$f(k) = 2k = t.$$

Hence f is onto.
Therefore f is an isomorphism, and $(\mathbb{Z}, +) \cong (E, +)$.

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